

EC602 — Computer Network

Semester VI • 3L:0T:0P • 3 Credits

Module 1 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1	Overview of Data Communication and Networking	2
1.1	Components of Data Communication	2
1.2	Data Representation	2
1.3	Direction of Data Flow	2
1.4	Network Criteria	3
2	Physical Structure: Topology and Categories	3
2.1	Network Topologies	3
2.2	Categories of Networks	4
3	Internet: History and Protocols	4
4	Reference Models	5
4.1	OSI Reference Model	5
4.2	TCP/IP Reference Model	5
4.3	OSI vs. TCP/IP — Comparison	6
5	Physical Level	6
5.1	Analog vs. Digital Data and Signals	6
5.2	Transmission Media	7
5.3	Circuit Switching	7
5.4	Telephone Network	8
6	Multiplexing: FDM, TDM, and WDM	8
6.1	Frequency Division Multiplexing (FDM)	8
6.2	Time Division Multiplexing (TDM)	9
6.3	Wavelength Division Multiplexing (WDM)	10
6.4	FDM vs. TDM vs. WDM — Comparison	12

1. Overview of Data Communication and Networking

Data Communication: The exchange of data between two devices via a transmission medium. It requires a *sender*, *receiver*, *message*, *medium*, and *protocol*.

1.1 Components of Data Communication

Component	Description
Message	The information (data) to be communicated — text, numbers, images, audio, video.
Sender	The device that sends the data message (computer, phone, camera).
Receiver	The device that receives the message (computer, printer, phone).
Medium	The physical path over which the message travels (cable, air, fibre).
Protocol	A set of rules governing data communication; ensures sender and receiver agree on format, timing, and error handling.

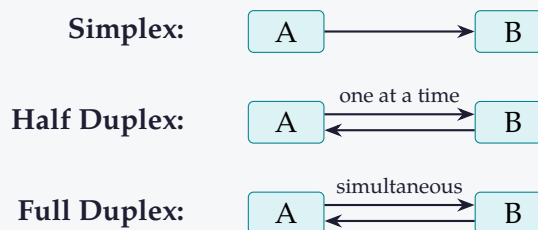
1.2 Data Representation

Data is represented in different formats depending on the type of information:

Type	Representation
Text	Encoded using bit patterns — ASCII (7-bit, 128 chars), ISO 8859 (8-bit, 256 chars), Unicode (32-bit, universal).
Numbers	Stored directly in binary; not converted to ASCII to save space.
Images	Represented as a matrix of pixels; each pixel encoded in bits (RGB).
Audio	Continuous signal sampled at regular intervals and quantised to binary values.
Video	Sequence of images (frames) displayed at a rate (fps) to create motion.

1.3 Direction of Data Flow

Data Flow Modes



Mode	Description	Example
Simplex	One direction only; receiver cannot send back.	TV broadcast, keyboard to CPU
Half Duplex	Both directions, but not simultaneously ; one at a time.	Walkie-talkie, CB radio
Full Duplex	Both directions simultaneously ; uses two separate channels or frequencies.	Telephone, mobile call

1.4 Network Criteria

A good network must satisfy three fundamental criteria:

Three Criteria of a Good Network

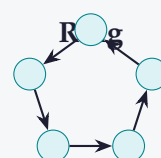
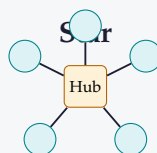
1. **Performance** — measured by transit time (time for a message to travel from sender to receiver) and response time. Affected by number of users, type of medium, hardware, and software.
2. **Reliability** — frequency of failure, recovery time after failure, robustness in catastrophe.
3. **Security** — protecting data from unauthorised access, damage, and implementing recovery policies.

2. Physical Structure: Topology and Categories

2.1 Network Topologies

Topology: The geometric arrangement of devices (nodes) and links in a network. It defines how devices are physically or logically connected.

Common Network Topologies



Topology	Description	Advantage / Disadvantage
Bus	All devices share a single backbone cable. Terminators at both ends.	Simple, cheap / Single point of failure; collisions.
Star	All devices connect to a central hub/switch.	Easy fault isolation / Hub failure = network down.
Ring	Each device connects to exactly two others forming a closed loop.	Equal access / One break disrupts entire ring.
Mesh	Every device connects to every other device (full mesh) or some (partial mesh).	Highly reliable / Expensive, complex wiring.
Tree (Hybrid)	Hierarchical combination of star and bus topologies.	Scalable / Backbone failure affects subtrees.

2.2 Categories of Networks

Category	Coverage	Description & Examples
LAN	Up to a few km	Local Area Network; single building or campus. Ethernet, Wi-Fi. High speed (100 Mbps–10 Gbps).
MAN	City-wide (5–50 km)	Metropolitan Area Network; connects multiple LANs in a city. Cable TV networks, WiMAX.
WAN	Country / Global	Wide Area Network; connects cities, countries. Internet, leased lines, satellite links.

3. Internet: History and Protocols

Internet: A global network of networks interconnected using the TCP/IP protocol suite. It evolved from ARPANET (1969), a US Department of Defense project for fault-tolerant communication.

Brief History:

- **1969** — ARPANET launched; first 4-node network (UCLA, Stanford, UCSB, Utah).
- **1974** — TCP/IP protocol defined by Cerf and Kahn.
- **1983** — ARPANET officially adopts TCP/IP; modern Internet born.
- **1991** — World Wide Web (WWW) introduced by Tim Berners-Lee at CERN.
- **1993** — Mosaic browser; Internet becomes publicly accessible.

Protocols and Standards: A **protocol** is a set of rules governing communication. A **standard** is a protocol agreed upon by industry bodies. Key bodies: **ISO, IEEE, IETF, ITU-T, ANSI**.

4. Reference Models

4.1 OSI Reference Model

OSI Model (Open Systems Interconnection): A 7-layer conceptual framework developed by ISO (1984) to standardise communication between different systems. Each layer has specific functions and communicates with adjacent layers via interfaces.

#	Layer	Function & PDU
7	Application	User interface; provides network services to applications. PDU: <i>Data</i> . Protocols: HTTP, FTP, SMTP, DNS.
6	Presentation	Translation, encryption, compression of data. PDU: <i>Data</i> .
5	Session	Establishes, manages, and terminates sessions between applications. PDU: <i>Data</i> .
4	Transport	End-to-end delivery, segmentation, flow control, error control. PDU: <i>Segment</i> . Protocols: TCP, UDP.
3	Network	Logical addressing (IP), routing, path determination. PDU: <i>Packet</i> . Protocols: IP, ICMP, RIP, OSPF.
2	Data Link	Framing, physical addressing (MAC), error detection, flow control. PDU: <i>Frame</i> . Protocols: Ethernet, HDLC.
1	Physical	Transmission of raw bits over medium; defines electrical/optical signals. PDU: <i>Bits</i> .

4.2 TCP/IP Reference Model

TCP/IP Model: A 4-layer practical model that forms the basis of the Internet. Developed by the DARPA. Less abstract than OSI; combines OSI layers 5, 6, 7 into Application and layers 1, 2 into Network Access.

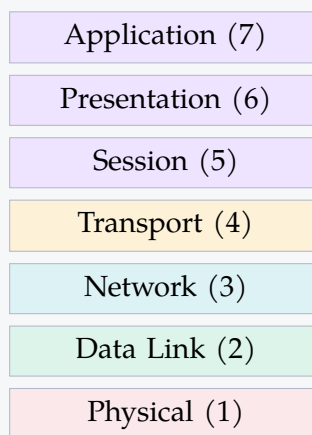
TCP/IP Layer	OSI Equivalent	Protocols
Application	Layers 5, 6, 7	HTTP, FTP, SMTP, DNS, SNMP, Telnet
Transport	Layer 4	TCP, UDP
Internet	Layer 3	IP (IPv4, IPv6), ICMP, ARP, RARP
Network Access	Layers 1, 2	Ethernet, Wi-Fi (802.11), Token Ring, Frame Relay

4.3 OSI vs. TCP/IP — Comparison

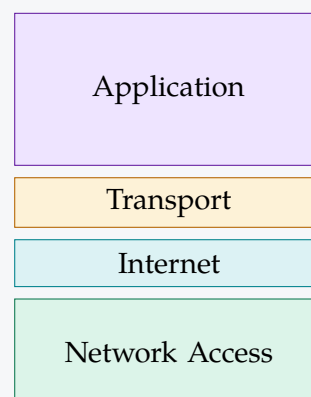
Parameter	OSI Model	TCP/IP Model
Layers	7 layers	4 layers
Developed by	ISO (1984)	DARPA (1970s)
Nature	Conceptual / theoretical	Practical / implemented
Transport layer	Connection & connectionless	Connectionless (UDP) + Connection (TCP)
Session/Presentation	Separate layers (5 & 6)	Merged into Application layer
Protocol dependency	Protocol-independent model	Protocol-specific (TCP/IP suite)
Usage	Reference / teaching tool	Actual Internet implementation

OSI vs. TCP/IP Layer Mapping

OSI (7 layers)



TCP/IP (4 layers)



5. Physical Level

5.1 Analog vs. Digital Data and Signals

Aspect	Analog	Digital
Data	Continuous values (e.g. voice, temperature)	Discrete values (0 and 1)
Signal	Continuous waveform (sine wave)	Square wave / pulse train
Noise immunity	Low — noise degrades signal	High — regenerated at repeaters
Bandwidth	Lower bandwidth needed	Higher bandwidth required
Transmission	Analog transmission (amplifiers)	Digital transmission (repeaters)

5.2 Transmission Media

Transmission Medium: The physical path between sender and receiver. Classified as **guided** (wired) or **unguided** (wireless).

Guided Media:

- **Twisted Pair Cable:** Two insulated copper wires twisted together. Reduces electromagnetic interference. Types: UTP (Unshielded) and STP (Shielded). Used in telephone lines, Ethernet (Cat5e, Cat6).
- **Coaxial Cable:** Central conductor surrounded by insulation, metallic shield, and outer jacket. Higher bandwidth than twisted pair. Used in cable TV, older Ethernet (10Base2, 10Base5).
- **Fibre Optic Cable:** Transmits data as light pulses through glass/plastic core. Extremely high bandwidth, immune to EMI, low attenuation. Used in backbone networks, long-distance links.

Unguided Media (Wireless):

- **Radio Waves:** Omnidirectional; penetrate walls. Used in AM/FM radio, Wi-Fi, Bluetooth.
- **Microwaves:** Directional; line-of-sight required. Used in satellite communication, cellular networks.
- **Infrared:** Short range; cannot penetrate walls. Used in TV remotes, IrDA.

5.3 Circuit Switching

Circuit Switching: A dedicated communication path is established between sender and receiver for the entire duration of the session. Resources are reserved end-to-end. Used in traditional telephone networks (PSTN).

Phases of Circuit Switching:

1. **Circuit Establishment:** A dedicated path is set up through intermediate switches.
2. **Data Transfer:** Data flows continuously along the reserved path.
3. **Circuit Disconnect:** Path is released and resources freed after communication ends.

Space Division Switch

Space Division Switching

Each connection uses a **physically separate path** through the switch matrix. A crosspoint (transistor/relay) is activated at the intersection of input and output lines.

Crosspoint matrix: For n inputs and n outputs, requires n^2 crosspoints.

Advantage: Simple, simultaneous connections. **Disadvantage:** Expensive for large n .

Time Division Switch (TDM Bus)

Time Division Switching

All connections share the **same physical medium** but are separated in **time slots**. A Time Slot Interchange (TSI) unit reads input slots and writes them to output slots.

TDM Bus: A high-speed shared bus; each input is assigned a time slot. The TSI reorders slots to route data to correct outputs.

Advantage: Efficient use of medium. **Disadvantage:** Requires synchronisation.

5.4 Telephone Network

The Public Switched Telephone Network (PSTN) uses circuit switching. It consists of:

- **Local loops:** Twisted pair from subscriber to local exchange (last mile).
- **Trunks:** High-capacity links (fibre/microwave) between exchanges.
- **Switching offices:** Local, tandem, and toll offices that route calls.

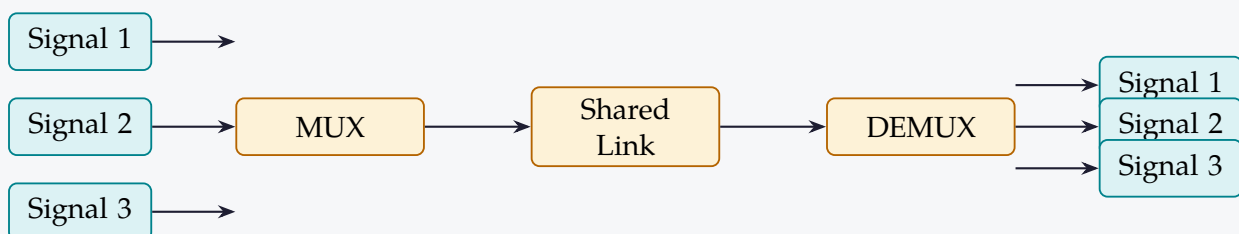
Telephone Network Structure



6. Multiplexing: FDM, TDM, and WDM

Multiplexing: A technique that allows multiple signals to share a single transmission medium simultaneously. A **multiplexer (MUX)** combines signals at the sender; a **demultiplexer (DEMUX)** separates them at the receiver.

General Multiplexing Concept



6.1 Frequency Division Multiplexing (FDM)

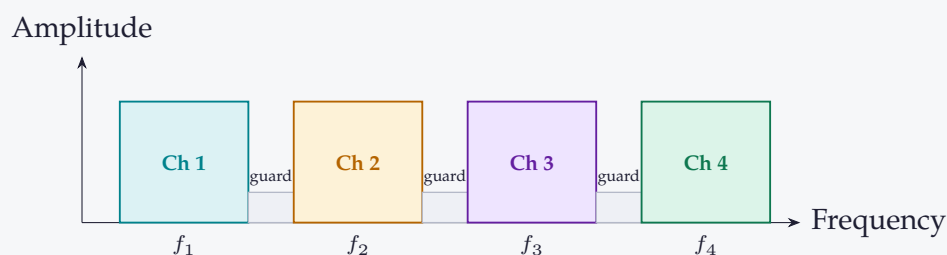
FDM (Frequency Division Multiplexing): The total bandwidth of the medium is divided into non-overlapping **frequency bands (sub-channels)**. Each signal is modulated onto a different carrier frequency and transmitted simultaneously. Used for **analog signals**.

Working:

1. Each input signal is modulated onto a unique carrier frequency $f_1, f_2, \dots, f_n$.
2. Guard bands (unused frequency gaps) separate adjacent channels to prevent interference.
3. All modulated signals are combined and transmitted over the shared medium.
4. At the receiver, bandpass filters separate each channel; demodulation recovers the original signal.

FDM Key Formulas

Channel bandwidth	$B_{ch} = f_{upper} - f_{lower}$
Total bandwidth	$B_{total} = n \times B_{ch} + (n - 1) \times B_{guard}$
Guard band	Unused frequency gap between adjacent channels to prevent inter-channel interference
Carrier frequencies	$f_1 < f_2 < \dots < f_n$, each separated by at least $B_{ch} + B_{guard}$

FDM — Frequency Spectrum Allocation

Aspect	Details
Signal type	Analog signals
Separation method	Different carrier frequencies; guard bands prevent overlap
Simultaneous?	Yes — all channels transmit at the same time
Applications	AM/FM radio, cable TV, ADSL, first-generation cellular (1G)
Advantage	Simple; no synchronisation needed; all channels always active
Disadvantage	Guard bands waste bandwidth; susceptible to intermodulation noise

6.2 Time Division Multiplexing (TDM)

TDM (Time Division Multiplexing): The time on the shared channel is divided into fixed **time slots**. Each input signal is assigned a slot in a repeating **frame**. Used for **digital signals**.

Two types of TDM:

Synchronous TDM

Each input is assigned a **fixed time slot** in every frame, regardless of whether it has data to send. Slots are pre-allocated.

Frame duration $T_f = n \times T_s$ (n = number of channels, T_s = slot duration)

Output bit rate $R_{out} = n \times R_{in}$ (output must be n times faster than each input)

Frame efficiency Can be low — empty slots wasted if a channel has no data

Statistical (Asynchronous) TDM

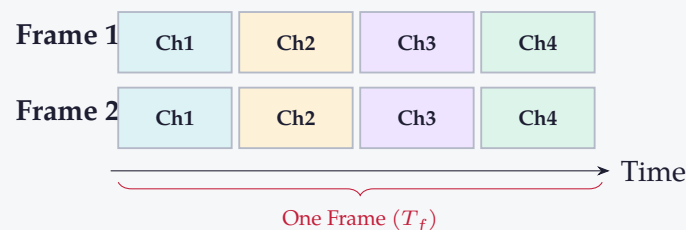
Slots are allocated **dynamically** — only to channels that have data. Each slot carries an **address header** to identify the source.

Efficiency Higher than synchronous TDM; no wasted slots

Overhead Address field added to each slot increases overhead

Output bit rate Less than $n \times R_{in}$ (statistical multiplexing gain)

Synchronous TDM — Frame Structure (4 channels)



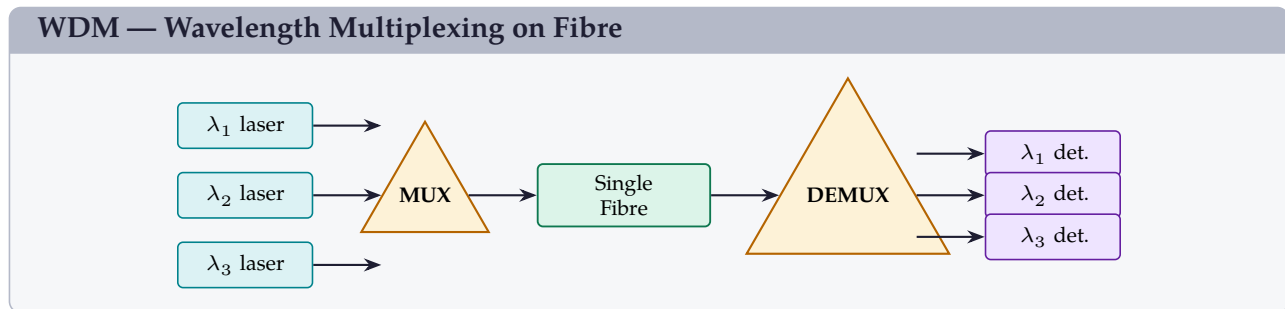
Aspect	Synchronous TDM	Statistical TDM
Slot allocation	Fixed; pre-assigned per channel	Dynamic; only active channels get slots
Address header	Not needed (position identifies channel)	Required (slot carries source address)
Efficiency	Low if channels idle	High; no wasted slots
Output rate	$n \times R_{in}$ (fixed)	Less than $n \times R_{in}$
Example	T1/E1 lines, SONET	ATM, packet-switched networks

6.3 Wavelength Division Multiplexing (WDM)

WDM (Wavelength Division Multiplexing): Multiple optical signals are transmitted simultaneously over a single **fibre optic cable**, each on a different **wavelength (colour) of light**. Conceptually identical to FDM but applied to optical frequencies. Used in high-capacity fibre networks.

Working:

1. Each data stream modulates a laser of a distinct wavelength $\lambda_1, \lambda_2, \dots, \lambda_n$.
2. A **prism or optical multiplexer** combines all wavelengths onto one fibre.
3. The combined beam travels through the fibre.
4. At the receiver, a **prism or optical demultiplexer** separates the wavelengths.
5. Each wavelength is detected by a photodetector and demodulated.

**Types of WDM:**

- **CWDM (Coarse WDM):** Fewer channels (up to 18), wider wavelength spacing (20 nm). Lower cost, shorter distances.
- **DWDM (Dense WDM):** Many channels (40–160+), very narrow spacing (0.8 nm or less). Used in long-haul backbone networks. Capacity up to several Tbps on one fibre.

Aspect	Details
Medium	Fibre optic cable only
Separation method	Different wavelengths (colours) of light
Analogy	FDM applied to optical domain
Capacity	DWDM: 40–160+ channels; Tbps aggregate
Applications	Internet backbone, long-haul telecom, submarine cables
Advantage	Enormous capacity; each wavelength is independent
Disadvantage	Expensive equipment; requires precise laser wavelengths

6.4 FDM vs. TDM vs. WDM — Comparison

Parameter	FDM	TDM	WDM
Full form	Frequency Division Multiplexing	Time Division Multiplexing	Wavelength Division Multiplexing
Signal type	Analog	Digital	Optical (light)
Sharing basis	Frequency bands	Time slots	Wavelengths of light
Medium	Coaxial, twisted pair, air	Any digital medium	Fibre optic only
Synchronisation	Not required	Required (sync TDM)	Not required
Guard bands	Required	Not needed	Not needed (wavelength spacing)
Applications	Radio, cable TV, ADSL	T1/E1, SONET, GSM	Internet backbone, DWDM
Capacity	Moderate	High	Very high (Tbps)

Quick Revision — Module 1

Data Communication, Networking Overview & Physical Level | EC602

Key Facts at a Glance

OSI Layers (top to bottom)	Application, Presentation, Session, Transport, Network, Data Link, Physical
TCP/IP Layers	Application, Transport, Internet, Network Access
LAN coverage	Up to a few km (building/campus)
MAN coverage	City-wide (5–50 km)
WAN coverage	Country / global
Simplex	One direction only
Half Duplex	Both directions, not simultaneously
Full Duplex	Both directions simultaneously
Circuit switching phases	Establish → Transfer → Disconnect
Space division crosspoints	n^2 for $n \times n$
TDM output bit rate	$R_{out} = n \times R_{in}$ (synchronous TDM)
TDM frame duration	$T_f = n \times T_s$
FDM total bandwidth	$B_{total} = n \times B_{ch} + (n - 1) \times B_{guard}$
WDM	FDM applied to optical domain; uses wavelengths of light switch
ASCII	7-bit encoding, 128 characters
Unicode	32-bit encoding, universal character set

One-Line Definitions

- **Protocol:** Set of rules governing communication between devices.
- **Topology:** Geometric arrangement of devices and links in a network.
- **LAN:** Local Area Network; high-speed network within a building or campus.
- **WAN:** Wide Area Network; spans cities, countries, or the globe.
- **OSI Model:** 7-layer ISO standard for network communication.
- **TCP/IP Model:** 4-layer practical model underlying the Internet.
- **Circuit Switching:** Dedicated path reserved for entire session duration.

- **TDM:** Time Division Multiplexing; shares medium by assigning time slots to each channel.
- **FDM:** Frequency Division Multiplexing; divides bandwidth into frequency bands for analog signals.
- **WDM:** Wavelength Division Multiplexing; uses different light wavelengths on a single fibre optic cable.
- **Statistical TDM:** Dynamic slot allocation; only active channels get slots, improving efficiency.
- **DWDM:** Dense WDM; 40–160+ channels with narrow wavelength spacing; used in backbone networks.
- **Guided Media:** Physical wired medium (twisted pair, coaxial, fibre optic).
- **Unguided Media:** Wireless medium (radio, microwave, infrared).
- **Full Duplex:** Simultaneous two-way communication.
- **ASCII:** American Standard Code for Information Interchange; 7-bit text encoding.

Frequently Asked / Viva Questions

Q1. What are the five components of data communication?

Sender, Receiver, Message, Medium, and Protocol.

Q2. Difference between simplex, half duplex, and full duplex?

Simplex: one direction only. Half duplex: both directions but not simultaneously. Full duplex: both directions simultaneously.

Q3. How many layers does the OSI model have? Name them.

7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application (bottom to top).

Q4. What is the PDU at the Network layer?

Packet.

Q5. What is the PDU at the Data Link layer?

Frame.

Q6. How does TCP/IP differ from OSI?

TCP/IP has 4 layers vs. OSI's 7; TCP/IP is practical and protocol-specific; OSI is a conceptual reference model.

Q7. What is circuit switching?

A dedicated communication path is established and reserved for the entire session. Used in PSTN.

Frequently Asked / Viva Questions (contd.)**Q8. What is the difference between space division and time division switching?**

Space division uses separate physical paths (crosspoint matrix); time division shares one medium using time slots (TSI).

Q9. What is the advantage of fibre optic over copper?

Higher bandwidth, lower attenuation, immune to EMI, more secure.

Q10. What is a MAN?

Metropolitan Area Network; covers a city (5–50 km); e.g. cable TV network, WiMAX.

Q11. What does ASCII stand for?

American Standard Code for Information Interchange; 7-bit encoding for 128 characters.

Q12. Why is full duplex preferred over half duplex?

Full duplex allows simultaneous two-way communication, doubling effective throughput compared to half duplex.

Q13. What is multiplexing and why is it used?

Multiplexing allows multiple signals to share a single transmission medium, reducing cost and making efficient use of bandwidth.

Q14. What is the difference between FDM and TDM?

FDM divides the bandwidth into frequency bands for analog signals; TDM divides time into slots for digital signals. FDM channels transmit simultaneously; TDM channels take turns.

Q15. What is WDM and where is it used?

WDM transmits multiple optical signals on one fibre using different wavelengths of light. Used in Internet backbone and long-haul telecom (DWDM supports 40–160+ channels).

Q16. What is the difference between synchronous and statistical TDM?

Synchronous TDM assigns fixed slots to each channel regardless of activity (wastes slots if idle). Statistical TDM allocates slots dynamically only to active channels, improving efficiency but requiring address headers.

OSI Layer Summary Table

#	Layer	Key Function	PDU
7	Application	User-facing services (HTTP, FTP, DNS)	Data
6	Presentation	Encryption, compression, translation	Data
5	Session	Session management, synchronisation	Data
4	Transport	End-to-end delivery, TCP/UDP	Segment
3	Network	Routing, IP addressing	Packet
2	Data Link	Framing, MAC, error detection	Frame
1	Physical	Bit transmission over medium	Bits